

3.2 Practice - Compound Inequalities

Solve each compound inequality, graph its solution, and give interval notation.

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|---|--|
| 1) $\frac{n}{3} \leq -3$ or $-5n \leq -10$ | 2) $6m \geq -24$ or $m - 7 < -12$ |
| 3) $x + 7 \geq 12$ or $9x < -45$ | 4) $10r > 0$ or $r - 5 < -12$ |
| 5) $x - 6 < -13$ or $6x \leq -60$ | 6) $9 + n < 2$ or $5n > 40$ |
| 7) $\frac{v}{8} > -1$ and $v - 2 < 1$ | 8) $-9x < 63$ and $\frac{x}{4} < 1$ |
| 9) $-8 + b < -3$ and $4b < 20$ | 10) $-6n \leq 12$ and $\frac{n}{3} \leq 2$ |
| 11) $a + 10 \geq 3$ and $8a \leq 48$ | 12) $-6 + v \geq 0$ and $2v > 4$ |
| 13) $3 \leq 9 + x \leq 7$ | 14) $0 \geq \frac{x}{9} \geq -1$ |
| 15) $11 < 8 + k \leq 12$ | 16) $-11 \leq n - 9 \leq -5$ |
| 17) $-3 < x - 1 < 1$ | 18) $1 \leq \frac{p}{8} \leq 0$ |
| 19) $-4 < 8 - 3m \leq 11$ | 20) $3 + 7r > 59$ or $-6r - 3 > 33$ |
| 21) $-16 \leq 2n - 10 \leq -22$ | 22) $-6 - 8x \geq -6$ or $2 + 10x > 82$ |
| 23) $-5b + 10 \leq 30$ and $7b + 2 \leq -40$ | 24) $n + 10 \geq 15$ or $4n - 5 < -1$ |
| 25) $3x - 9 < 2x + 10$ and $5 + 7x \leq 10x - 10$ | 26) $4n + 8 < 3n - 6$ or $10n - 8 \geq 9 + 9n$ |
| 27) $-8 - 6v \leq 8 - 8v$ and $7v + 9 \leq 6 + 10v$ | 28) $5 - 2a \geq 2a + 1$ or $10a - 10 \geq 9a + 9$ |
| 29) $1 + 5k \leq 7k - 3$ or $k - 10 > 2k + 10$ | 30) $8 - 10r \leq 8 + 4r$ or $-6 + 8r < 2 + 8r$ |
| 31) $2x + 9 \geq 10x + 1$ and $3x - 2 < 7x + 2$ | |
| 32) $-9m + 2 < -10 - 6m$ or $-m + 5 \geq 10 + 4m$ | |